Instagram Influencer Verification System

Overview

> [!IMPORTANT] **Disclaimer:**
>
> ❗ **NDA Protected:** This repository **DOES NOT** contain source code or runnable binaries due to Non-Disclosure Agreements with the client.
>
> ❗ **Academic Context:** This project was developed as a Capstone Software Project at the University of Melbourne, in collaboration with Dr. Brent Coker.
>
> ❗ **Status:** The system is a production-ready prototype delivered to the client for future integration.
>
> ❗ **Data Privacy:** All user data and metrics shown in screenshots are either **synthetic/mock data** or sourced from **publicly available profiles** (e.g., public figures) for demonstration purposes only. No private client data or real applicant information is disclosed.

We developed an **AI-driven Instagram Influencer Verification System** for **Wear Cape**. Designed to integrate seamlessly into the influencer registration workflow, this tool automates the assessment of account performance, generating a comprehensive quality score (0–100) and identifying potential red flags. By leveraging advanced LLMs and heuristic analysis, it provides actionable AI insights, significantly streamlining the vetting process for admin users.

The Problem

With the booming creator economy, brands and influencers need a reliable platform to connect and collaborate. **Wear Cape** was built to bridge this gap.

However, a critical bottleneck remained: **Verification**.

- **Data Scarcity:** Official Instagram APIs are restrictive and difficult to obtain for startups.
- **Manual Bottleneck:** Admins had to manually verify screenshots and metrics for every applicant.
- **Inefficiency:** The manual review process was estimated to take **30+ minutes per application**, making scaling impossible.

The Solution

Our team engineered a **secure, scalable, and automated AI pipeline**. The system extracts performance data from user-uploaded screenshots via OCR, analyzes it against heuristic rules, and uses LLMs to provide search-grounded, real-life context.

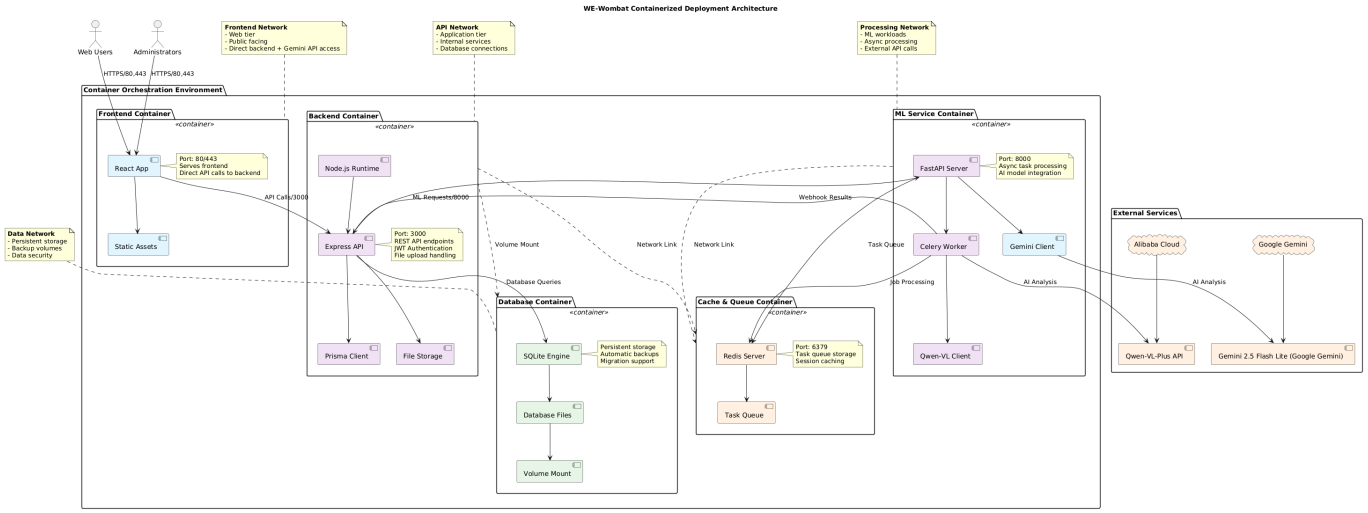
Impact: Reduced verification time from **~30 minutes** to **< 2 minutes** per application, achieving a **15x efficiency gain**.

My Contributions (Tech Lead)

As the **Technical Lead**, I orchestrated the end-to-end development lifecycle, from architectural design to final deployment, contributing **~75% of the production codebase**.

System Architecture & Design

- **Architecture & UML Design:** Architected a decoupled microservices system. Produced comprehensive UML diagrams, sequence diagrams, and API documentation to guide the engineering team and ensure independent scalability.
- **Microservices Architecture:** Architected a decoupled system separating the Frontend, Backend, and ML services to ensure independent scalability.
- **Data Integrity:** Designed a robust Database Schema (Prisma/SQLite/SQL) featuring a **Snapshot Pattern** to preserve historical applicant data against updates.
- **Security Protocols:** Established comprehensive security standards for authentication (JWT), file uploads, and API communications to mitigate OWASP Top 10 vulnerabilities.

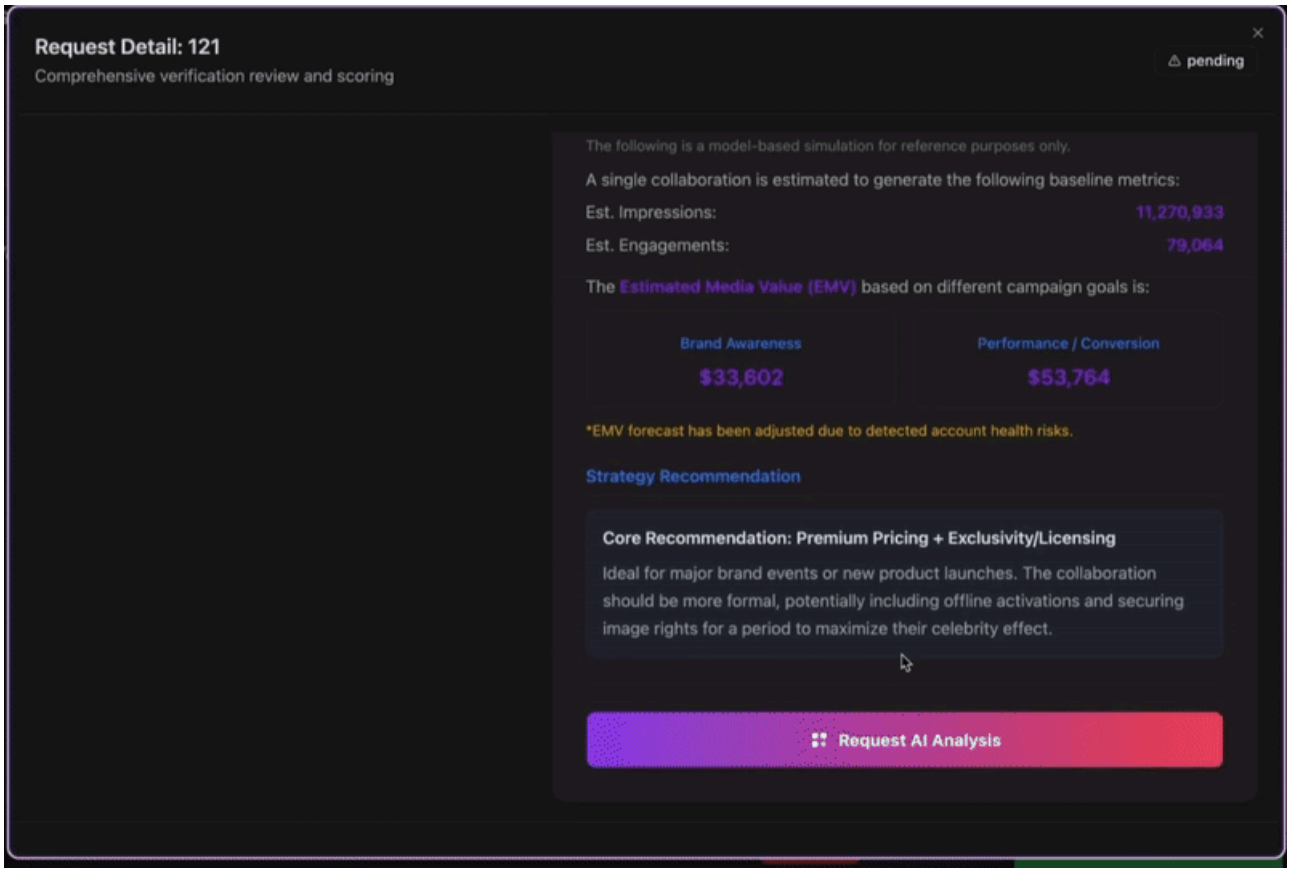


Backend Engineering (Node.js + Express)

- **Core Infrastructure:** Built the RESTful API backbone using **Node.js, Express, and TypeScript**, ensuring type safety and maintainability.
- **Secure Authentication:** Implemented a stateless authentication system using JWT with strict CORS policies and rate limiting to prevent Broken Access Control.
- **Async Processing:** Engineered a webhook-based communication layer to handle long-running ML inference tasks asynchronously.
- **ORM Integration:** Utilized **Prisma** for efficient database queries and migrations.

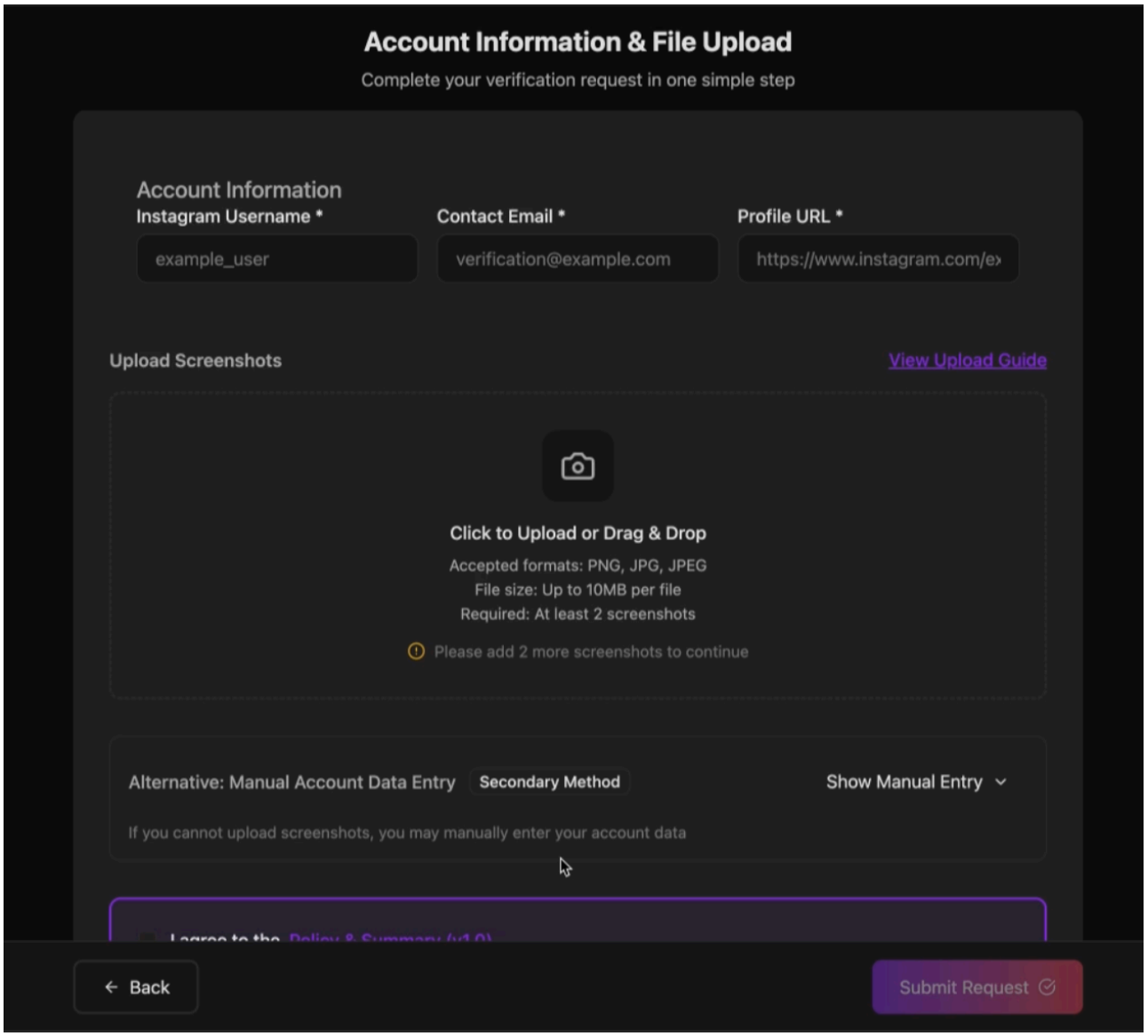
AI & Machine Learning Infrastructure

- **Scalable ML Pipeline:** Engineered the ML service using **Python (FastAPI)**.
- **Distributed Task Queue:** Implemented **Celery + Redis** to manage asynchronous inference tasks, preventing request timeouts and ensuring system responsiveness under load.
- **LLM Integration:** Integrated **Qwen-VL-Plus** and **Gemini** models for "Deep Dive" analysis.
- **Prompt Engineering:** Refined prompts to minimize hallucination and maximize information density in generated influencer reports.



Frontend Development (React + TypeScript)

- **Interactive UI:** Developed the "AI Deep Dive" interface and real-time status visualizations using **React 18**.
- **Data Visualization:** Implemented dynamic charts to render influencer metrics and engagement trends.
- **State Management:** Consolidated complex frontend logic to handle asynchronous data fetching and real-time updates from the backend.


Account Information & File Upload
Complete your verification request in one simple step
Account Information
Instagram Username *
Contact Email *
Profile URL *
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verification@example.com
https://www.instagram.com/e›
Upload Screenshots
View Upload Guide
Click to Upload or Drag & Drop
Accepted formats: PNG, JPG, JPEG
File size: Up to 10MB per file
Required: At least 2 screenshots
Please add 2 more screenshots to continue
Alternative: Manual Account Data Entry
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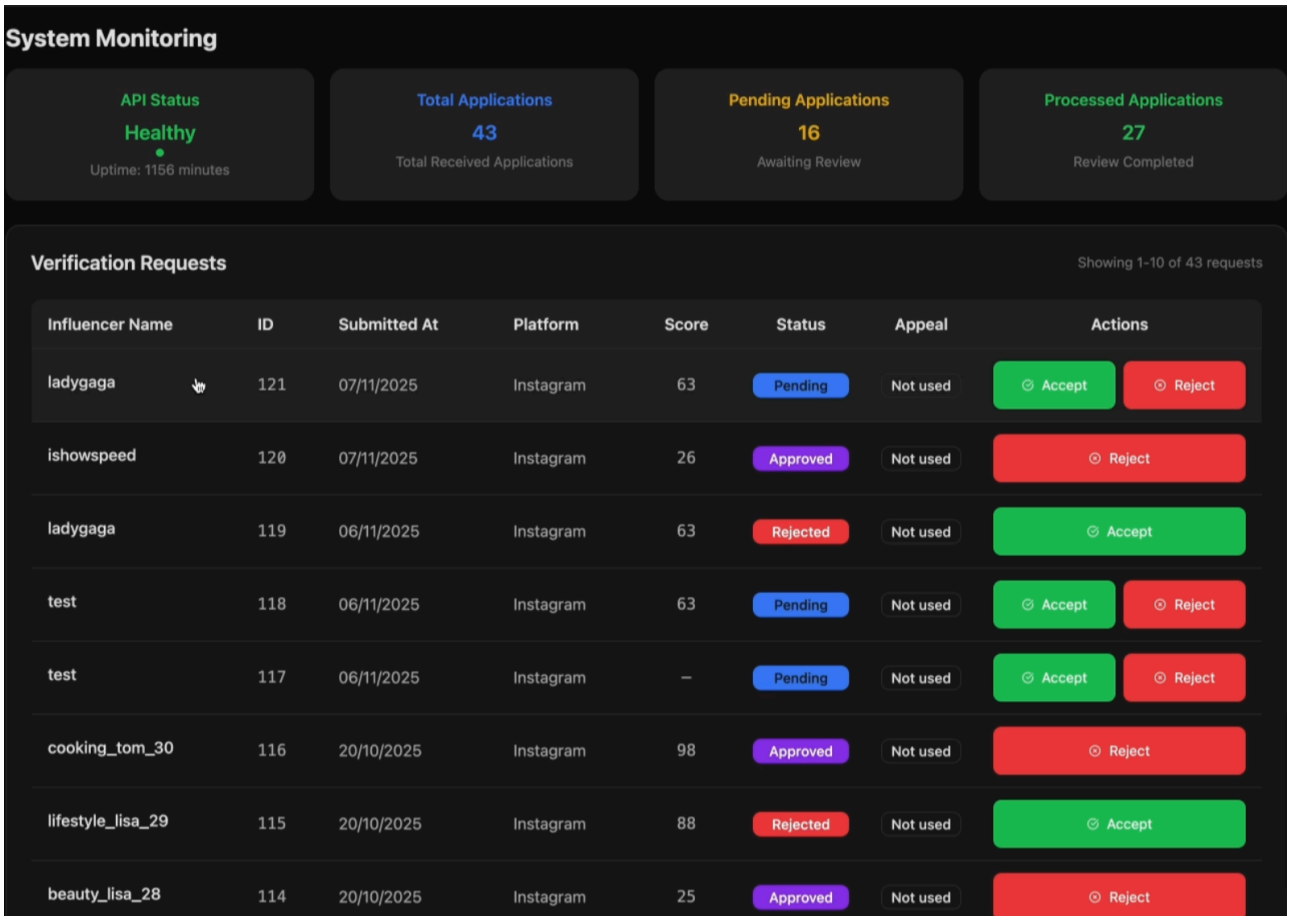
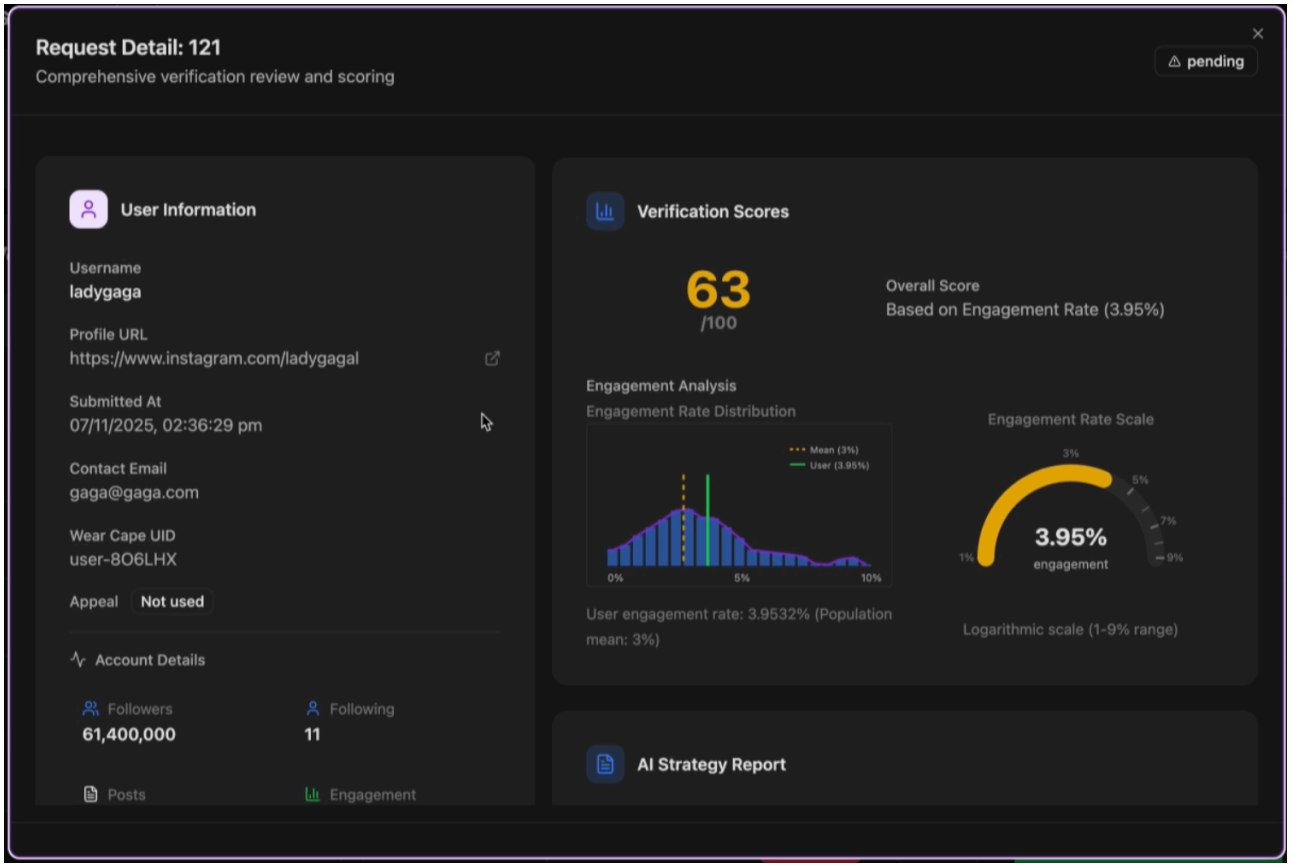


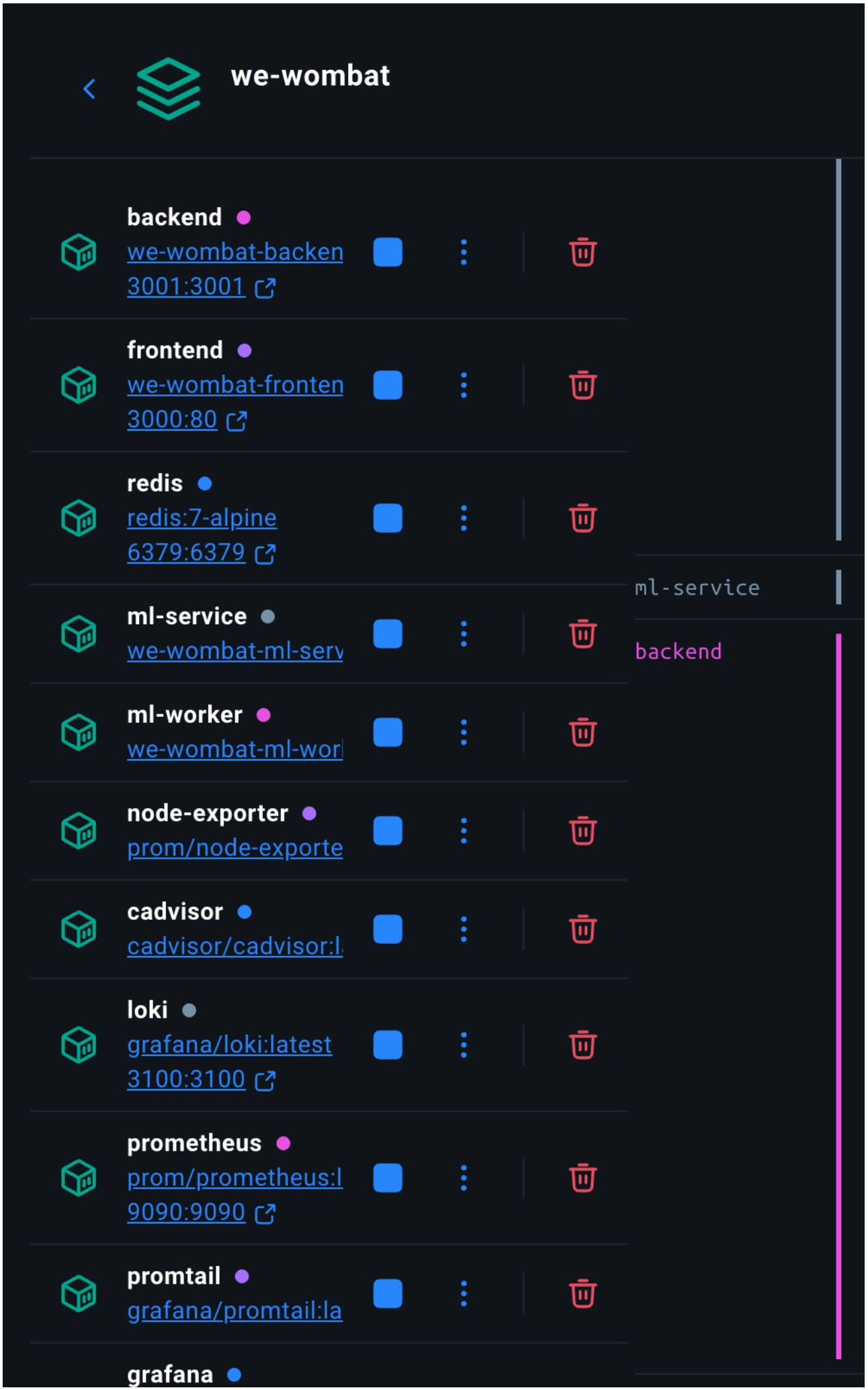


DevOps, CI/CD & Observability

- **Containerization:** Dockerized the entire stack (Frontend, Backend, ML Service, Workers, Redis) using **Docker Compose** for consistent deployment across environments.

- **Full Observability Stack:** Deployed a complete monitoring solution using **Prometheus** (metrics), **Grafana** (visualization), **Loki** (logs), and **Promtail**, ensuring production-grade visibility.
- **Automated Testing:** Established a rigorous testing culture and formulated comprehensive test plans using **Jest**. Achieved **>85% statement coverage** (1018/1195 statements) with 778+ test cases, ensuring high engineering standards.

- **CI/CD Pipelines:** Configured GitHub Actions for automated testing, linting, and build verification.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backend
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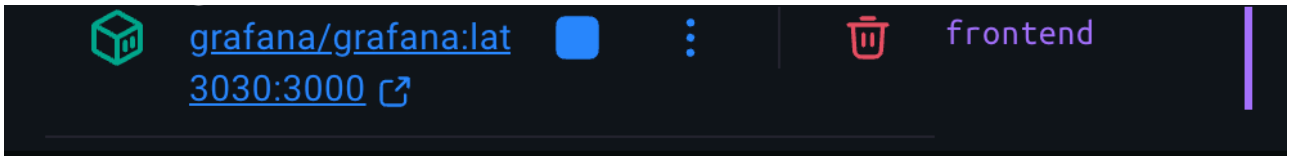
backend



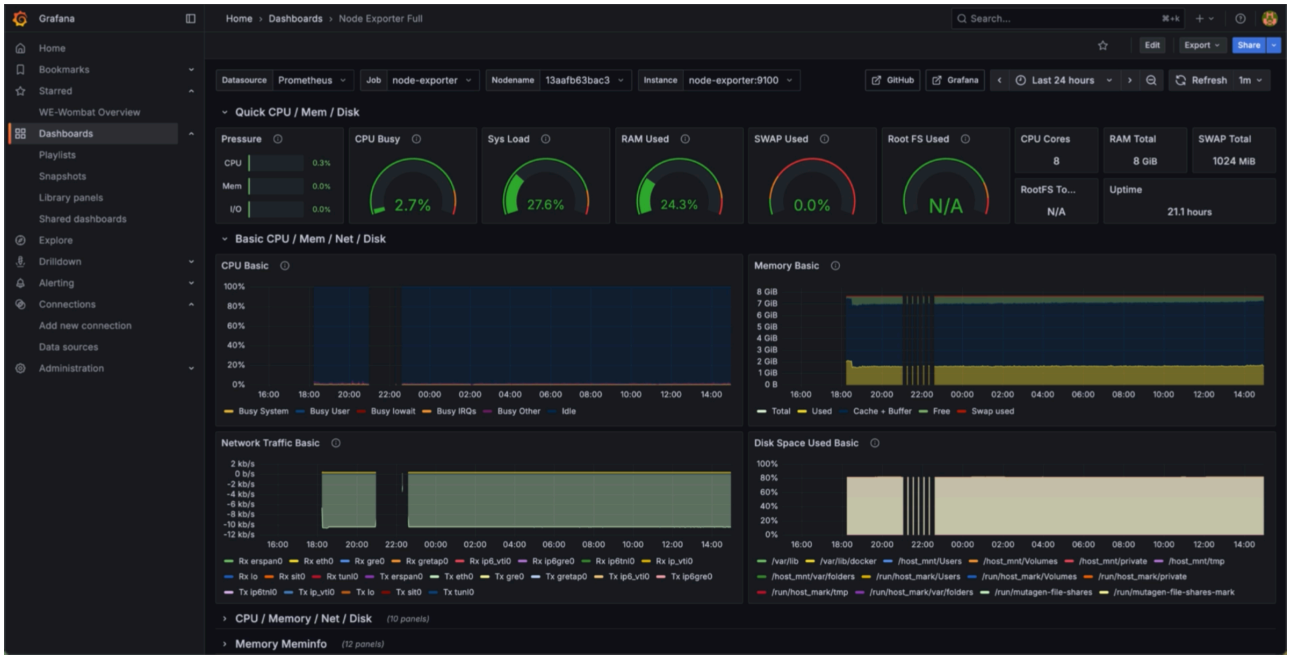
All files

85.18% Statements `1018/1195` **79.07%** Branches `514/650` **87.74%** Functions `136/155` **85.22%** Lines `1004/1178`

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

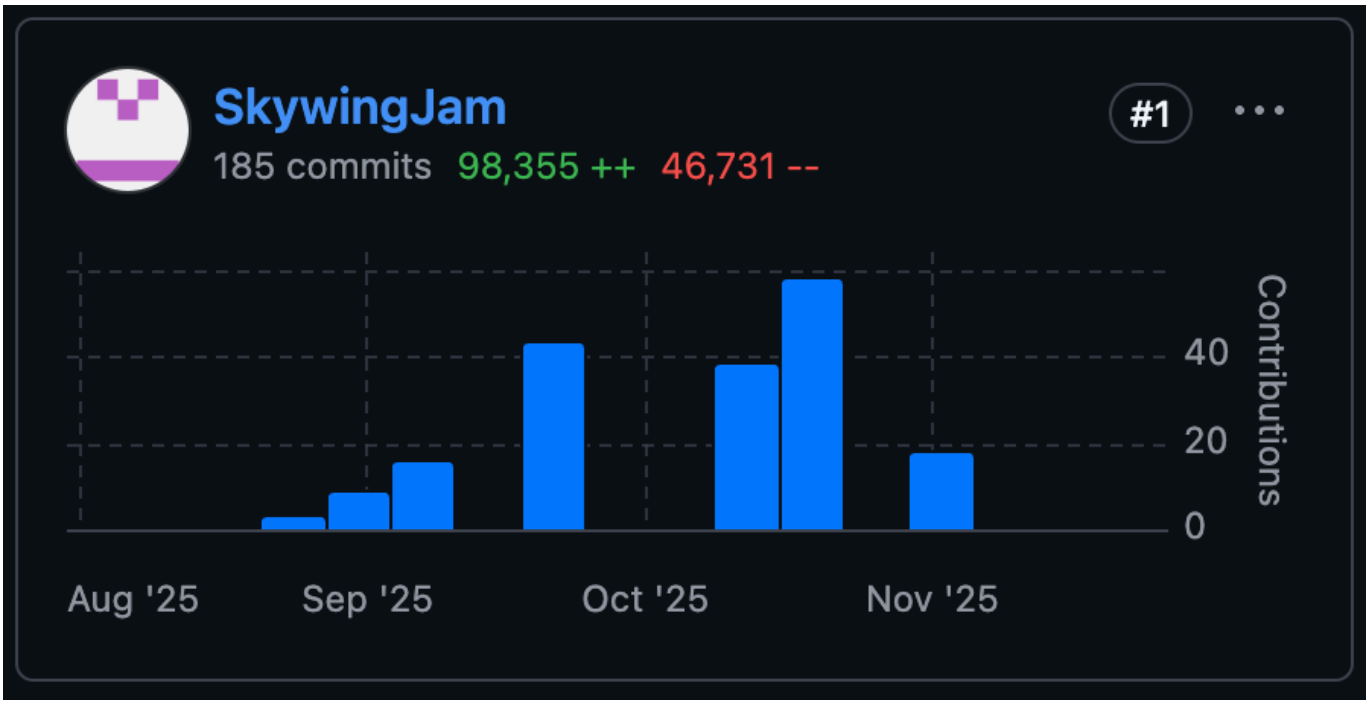
File		Statements		Branches		Functions		Lines	
src		0%	0/42	0%	0/8	0%	0/4	0%	0/42
src/controllers		83.65%	471/563	81.21%	294/362	82.75%	48/58	83.77%	470/561
src/middleware		100%	114/114	98%	49/50	100%	9/9	100%	106/106
src/routes		100%	88/88	91.66%	11/12	100%	5/5	100%	88/88
src/services		88.91%	345/388	73.39%	160/218	93.67%	74/79	89.23%	340/381



Agile Leadership

- **Technical Direction:** Bridged the gap between client business requirements and technical feasibility.
- **Sprint Management:** Managed task dependencies and technical debt, ensuring the team delivered a production-ready MVP despite tight deadlines.
- **Code Quality:** Enforced strict code review standards and Git workflows to maintain codebase health.

Development Logs



Outcome

The system received high praise from both the University teaching team and the client for its architectural maturity and completeness.

Final Presentation Instructions Final product/presentation	Oct 24 by 17:00	Oct 24 at 12:31	8 / 8

Final Presentations

Criteria	Ratings			Pts
Slide Design & Visual Impact view longer description	**2 to >1.5 pts** **Excellent** The slides are well-structured, professionally designed, and visually appealing. Typography, colors, and visual elements are consistent and enhance readability. The slides effectively support the presentation without excessive text. Final product scenarios are well-illustrated with supporting diagrams or UI previews.	**1.5 to >0.5 pts** **Good** The slides are mostly clear and visually consistent but may have minor design issues, such as too much text or inconsistent formatting. Final product scenarios are present but could be better illustrated.	**0.5 to >0 pts** **Inadequate** The slides are cluttered, unstructured, or inconsistent in design. Readability is poor due to small fonts, excessive text, or weak contrast. Final product is not well-illustrated, making it hard to understand.	2 / 2 pts
	Comments IMPRESSIVE!!! Love the slide and the theme colour matching the Instagram.			
Organisation & Delivery view longer description	**2 to >1.5 pts** **Excellent** The presentation is well-rehearsed and delivered confidently. Transitions between speakers are smooth, and all team members actively contribute. The structure is clear and easy to follow. The presenters engage the audience with clear explanations and appropriate pacing.	**1.5 to >0.5 pts** **Good** The presentation is mostly clear but could be more polished. Some transitions between speakers are slightly uncoordinated but do not significantly disrupt flow. Engagement is moderate, but explanations could be more refined.	**0.5 to >0 pts** **Inadequate** The presentation is unstructured or difficult to follow. Transitions between speakers are disjointed. Lack of engagement, poor pacing, or excessive reliance on reading from slides reduces effectiveness.	2 / 2 pts
	Comments AWESOME! The transitions between each speakers are smooth. Well delivered.			
Final Product Demonstration & Realistic Scenarios view longer description	**4 to >3 pts** **Excellent** The final product is presented clearly and professionally, with a structured demo. The scenarios are realistic, well-explained, and aligned with project goals. The demonstration focuses on key functionalities without unnecessary distractions. Requirements developed throughout the project are incorporated where relevant.	**3 to >1.5 pts** **Good** The final product is presented, but the demonstration lacks depth or clarity. The scenarios are present but could be more refined. Some key functionalities are shown, but not all are well-explained.	**1.5 to >0 pts** **Inadequate** The demonstration is unclear, lacks structure, or is incomplete. The scenarios are unrealistic or poorly connected to project goals. The final product is not effectively demonstrated.	4 / 4 pts
	Comments VERY ENGAGING! The role play for the scenarios is really interesting. It showcases the main features and the product itself looks amazing. Well done!			
				Total Points: 8

You can check out the final presentation recording here (Click on the Thumbnail to watch on YouTube), I'm the last speaker:

Tech Stack

- **Frontend:** React, TypeScript, Vite, TailwindCSS
- **Backend:** Node.js, Express, Prisma, SQLite
- **ML/AI:** Python, FastAPI, Celery, Redis, Qwen-VL/Gemini
- **DevOps:** Docker, GitHub Actions, Prometheus, Grafana, Loki

License & Copyright

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Credits

Special thanks to **Dr. Brent Coker** (Client), **Dr. Michael Fu** (Supervisor), and **Mr. Jim Hsiao** (Tutor) for their continuous support.